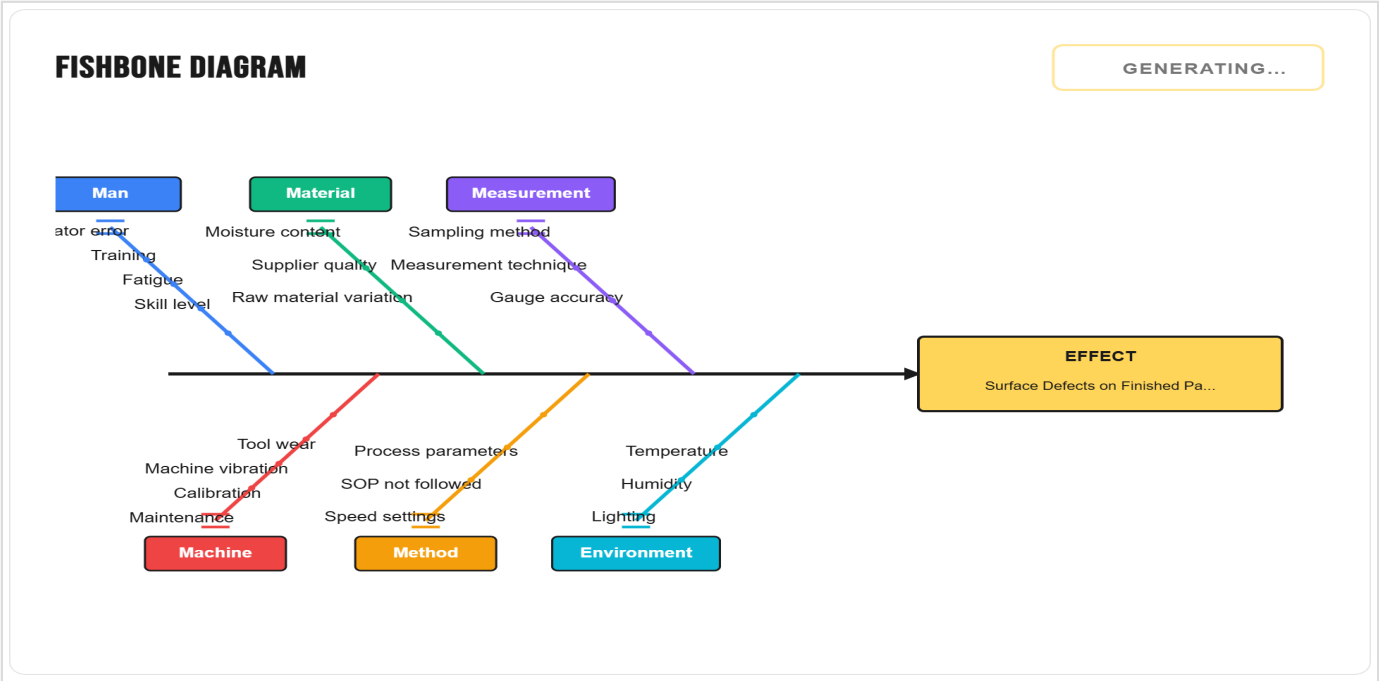


Cause & Effect (Fishbone) Diagram Report

Fishbone analysis for "Surface Defects on Finished Parts" identifies 20 potential root causes with 13 subcauses across 6 categories (Man: 4, Machine: 4, Material: 3, Method: 3, Measurement: 3, Environment: 3).



Ishikawa fishbone diagram — 6 categories, 20 causes, 13 subcauses for: "Surface Defects on Finished Parts"

PROBLEM STATEMENT (EFFECT)

Surface Defects on Finished Parts

ANALYSIS OVERVIEW

Effect / Problem Surface Defects on Finished	Categories Analysed 6
Total Causes 20	Total Subcauses 13
Most Causes In Man (4)	

CATEGORY SUMMARY

Category	Causes	Subcauses	Total Factors
Man	4	5	9
Machine	4	3	7
Material	3	1	4
Method	3	1	4
Measurement	3	1	4
Environment	3	2	5

MAN — CAUSES & SUBCAUSES

- Skill level ! Lack training, New hires

- Fatigue !' Long shifts
- Training
- Operator error !' Distraction, Rushing

MACHINE — CAUSES & SUBCAUSES

- Tool wear !' Old blades
- Machine vibration
- Calibration !' Not checked
- Maintenance !' Delayed

MATERIAL — CAUSES & SUBCAUSES

- Raw material variation
- Supplier quality !' Different batch
- Moisture content

METHOD — CAUSES & SUBCAUSES

- Process parameters !' Speed too high
- SOP not followed
- Speed settings

MEASUREMENT — CAUSES & SUBCAUSES

- Gauge accuracy !' Need calibration
- Measurement technique
- Sampling method

ENVIRONMENT — CAUSES & SUBCAUSES

- Temperature !' Too hot
- Humidity
- Lighting !' Poor visibility

ISHIKAWA'S METHODOLOGY

The cause closest to the effect on the diagram is not necessarily the root cause. Trace back through the bones to find the true root cause. Subcauses help drill down into specific contributing factors. Use data (Check Sheets, Pareto) to determine which subcauses have the highest impact.

RECOMMENDATIONS & ACTION PLAN

- 1 Investigate the highest-density categories first (Man has the most causes)
- 2 Use Check Sheets to quantify the frequency of each identified cause
- 3 Create a Pareto Chart to rank causes by their actual impact / cost

- 4 Drill down into the top 3 causes using the "5 Whys" technique
- 5 Develop targeted corrective action plans for confirmed root causes